Cyclistic Bike-Share Case Study

Prepared by: Naveenkumar K
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1. Ask

The objective of this analysis is to understand how casual riders and members use Cyclistic bikes differently. The business goal is to identify key behavioral patterns and provide actionable insights to help convert casual riders into annual members.

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2. Prepare

The dataset consists of 12 months of Cyclistic historical trip data. It includes variables such as ride ID, ride type, start and end times, and user type (casual or member). The data is publicly available and provided by Motivate International Inc.

A total of approximately 1 million rides were analyzed for this study.

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3. Process

Data cleaning and preparation were performed using Microsoft Excel and Power Query.

The following steps were taken:

- Combined multiple monthly datasets into a single dataset
- Removed unnecessary columns and handled missing values
- Created a new column **ride_length** by calculating the difference between start and end times
- Converted ride duration into minutes (**ride_length_min**)
- Extracted the **day of the week** from ride start time

These steps ensured the dataset was clean and ready for analysis.

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4. Analyze

◇ Average Ride Duration

Casual riders take longer rides compared to members.

- Casual Users: ~23.6 minutes
- Members: ~12 minutes

This indicates that casual riders are more likely using bikes for leisure purposes.

◇ Ride Frequency

Members account for a higher number of rides.

- Members: ~65% of total rides
- Casual Users: ~35% of total rides

This suggests that members are more frequent and consistent users.

◇ Weekly Usage Patterns

There is a clear difference in usage behavior:

- Members ride more during **weekdays (Monday–Friday)**
- Casual users ride more during **weekends (Saturday–Sunday)**

This shows that members primarily use bikes for commuting, while casual users use them for recreational purposes.

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5. Share

The analysis was visualized using pivot tables and charts in Excel.

The following visualizations were created:

- Average ride duration by user type
- Total rides by user type
- Daily ride distribution by user type

These visualizations clearly highlight the behavioral differences between casual riders and members.

6. Act

Based on the analysis, the following recommendations are proposed:

- Offer **weekend membership promotions** targeting casual riders
- Highlight **cost savings of annual membership** for frequent riders
- Launch **targeted digital marketing campaigns** during peak casual usage times (weekends)
- Introduce **loyalty programs or incentives** to encourage repeat usage

Conclusion

The analysis reveals clear behavioral differences between casual riders and members:

- Members are frequent users with shorter, weekday rides
- Casual users take longer rides and are more active on weekends

By targeting casual users with tailored marketing strategies, Cyclistic can effectively increase annual memberships and improve customer retention.